

S V V Sai Sri Vallabh Pataneni

Renningen / Stuttgart · vallabh.pataneni@rwth-aachen.de · +49 176 8594 4047 · [linkedin.com/in/vallabh-pataneni](https://www.linkedin.com/in/vallabh-pataneni)

Institute of Climate and Energy Systems (ICE-2), Jülicher Systemanalyse

Forschungszentrum Jülich

12 June 2026

Application: Masterarbeit, Importpotenziale grüner Wasserstoffderivate

I am applying for the Master's thesis on the import potential of green hydrogen derivatives. The integrated simulation and optimisation approach your institute uses is the way I already work: at Robert Bosch I built MATLAB/Simulink models of fuel-cell powertrain strategies and ran multi-variable optimisation with Genetic Algorithms and Gradient Descent, benchmarked against dynamic programming and ECMS, to find cost-effective operating points.

The techno-economic side of the thesis fits my background too. I built a Well-to-Wheel and total-cost-of-ownership model comparing powertrains across their life cycle, which is the same kind of system-level, cost-driven comparison that an import-potential analysis of hydrogen carriers requires. I am fluent in Python and MATLAB for the data handling and scenario analysis behind such models, and I follow the hydrogen literature with genuine interest through my fuel-cell work at DLR and Bosch.

Contributing to Jülich's work on a sustainable hydrogen economy is exactly the direction I want my thesis to take. My English is fluent and my German is at B1. I would be glad to discuss the scope with you.

Yours sincerely,

Vallabh Pataneni